ANAND RAMNARAIN

Software Engineer — C++, systems and full-stack development
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Early-career software engineer completing a Master's in Advanced Programming for AAA Games at UPC Barcelona (finishing November 2026). Experienced in C++ engine and library development, cross-platform rendering, resource management and automated testing, with full-stack experience in TypeScript, React and PostgreSQL. I enjoy tracing problems to their root cause and building reliable tools for other developers.

PROJECTS

Phoenix Engine — *C++20 / DirectX 12 engine* Jan 2026 – present · source

- **Independent development:** expanded the original Master's coursework model viewer (v0.2) into a standalone engine, independently releasing versions 0.3–0.5. Added typed ECS-style components, JSON scenes and prefabs, hot-reloaded C++ gameplay scripts and standalone game builds. The engine now powers my game Ashfall Keep.
- **Systems:** deferred lighting with compute-shader light culling, shadows, GPU skinning and particles; collision with three swappable broad phases (brute force, uniform grid, octree), impulse physics and A* pathfinding.
- **Debugging:** fixed a GPU crash from freeing resources still in use, and broken skinned-model imports (matrix order); cut per-draw overhead by caching descriptor tables per material.

Arclight — *cross-platform C++20 engine and editor for Windows and macOS* Oct 2026 – present · source

- **Rendering architecture:** designed a shared rendering interface for DirectX 12 and Metal, alongside a GPU-free back end for testing. Shaders are authored once in HLSL and compiled for both platforms at build time.
- **Reliability:** implemented generational handles and fence-based deferred resource release; added bounds-checked glTF import, 78 unit tests and GitHub Actions CI for both platforms.

Cross-Engine Spatial Data Streaming SDK — *C++20 library* 2026 · source

- Designed one API that streams large tiled 3D datasets within a fixed memory budget (quadtree, level of detail, background loading) for Unity, Unreal and a DirectX 12 engine; 187 automated tests. In a stress test, memory stayed at 1–3 MB as the world grew 16×, against 268 MB when loading everything.

Team Formation Platform — *personal full-stack web project* Dec 2024 – Mar 2025 · source

- React and TypeScript app on a serverless TypeScript API (Azure Functions) and PostgreSQL via Prisma: students vote on project ideas, winning ideas become teams and everyone else is assigned automatically. Docker, OpenAPI.

Bound by Death — *co-op action game, 17-person UPC Master's team* 2025 – 2026 · site

- On our C++ / DirectX 12 engine I built the prefab system, light culling (checked with PIX), post-processing, a JSON-driven controller-vibration system designers tune without code, and VFX for both playable characters. Scrum sprints and code reviews in both directions.

EXPERIENCE

Software Developer (vacation work) — *BBD Software* Nov 2024 – Dec 2024

- Built a team web application using React, TypeScript, PostgreSQL, Prisma and Flyway migrations, with Docker and Azure supporting development and testing.

Web Developer (freelance) — *Easy Shoe & Bag Repair* Jan 2023 – Nov 2023

- Built and maintained the business's website, product catalogue and search optimisation, working directly with the owner on requirements.

EDUCATION

Master's in Advanced Programming for AAA Games — *UPC, Barcelona* Oct 2025 – Nov 2026 (expected)

- C++, DirectX 12, engine architecture, graphics, animation, AI and physics; team game project in Scrum.

BA Digital Arts (Honours) — *University of the Witwatersrand, Johannesburg* Jan 2021 – Dec 2024

- Game Design and Interactive Media: Unity and C#, JavaScript and React; final-year research project on games and AI.

SKILLS

Programming languages: C++17/20, C#, TypeScript, JavaScript, SQL, HLSL
Engineering: OOP and component design, API design, data structures and algorithms, memory management, debugging
Testing and tools: unit testing (doctest, CTest, NUnit), CMake, GitHub Actions CI, Git, Visual Studio, PIX, RenderDoc
Web and back end: React, Node.js, Azure Functions, PostgreSQL, Prisma, Flyway, Docker, Azure, OpenAPI
Graphics and engines: DirectX 12, Metal, real-time rendering, compute shaders, Unity, Three.js, Unreal (coursework)
Spoken languages: English (fluent), Spanish (beginner, currently learning)